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# UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

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## ABSTRACT

The computational complexity of deriving two fundamental types of explanations for ML model predictions — (i) *sufficient reasons*, subsets of input features that, when fixed, determine a prediction, and (ii) *contrastive reasons*, subsets of input features that, when modified, alter a prediction— has so far been studied only piecemeal, in separate contexts such as non-probabilistic versus probabilistic frameworks and local versus global settings. This paper makes four contributions that together give a unified account of this landscape. **First**, we introduce a single framework that recasts *all* of these explanation problems as the minimization of one probabilistic value function, and we isolate three properties of that value function that govern its complexity: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — three cornerstone properties of *combinatorial optimization*. **Second**, using this framework we uncover a counterintuitive dichotomy: whereas the *local* value functions possess *none* of these properties, we prove that their *global* counterparts possess *all* of them. **Third**, we turn this structural dichotomy into a suite of new polynomial-time algorithms that compute global explanations with provable guarantees across models spanning the interpretability spectrum — neural networks, decision trees, and tree ensembles. **Fourth**, we show the gap is fundamental: even drastically simplified versions of these same explanations are NP-hard to compute in the corresponding local setting. Taken together, our results explain *why* global explanations can be tractable precisely where local ones are not.

## 1 INTRODUCTION

Despite substantial progress in methods for explaining ML model decisions, the literature has repeatedly reached a discouraging conclusion: many of the most desirable explanation types, precisely those that come with provable guarantees, are computationally hard to obtain (Barceló et al., 2020; Van den Broeck et al., 2022), and the difficulty typically worsens for the complex, highly non-linear models where explanations matter most (Barceló et al., 2020; Adolfi et al., 2025; 2024). This paper challenges the assumption that such hardness is unavoidable, by identifying a broad and practically relevant setting in which guaranteed explanations become tractable. Accordingly, the *computational complexity* of obtaining explanations has become a central theoretical focus, with a large body of recent work charting which explanation types can be computed efficiently for which models, and which remain out of reach (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022). Our contribution to this map is to reveal a systematic reason behind many of its tractable and intractable regions at once.

**From sufficient to contrastive reasons.** Among studies on the computational complexity of generating explanations, two fundamental types of explanations for ML models were extensively examined: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features  $S$  such that when these features are fixed to specific values, the model’s prediction remains unchanged, regardless of the values assigned to the complementary set  $\bar{S}$ . A *contrastive reason* is a subset of input features  $S$  such that modifying these features leads to a change in the model’s prediction.

These two explanation types are the focus of our study, and it is worth recalling why they matter. Unlike additive attribution methods, which allocate importance scores across features but are often hard for humans to interpret (Kumar et al., 2020) or lack actionability (Bilodeau et al., 2024), sufficient and contrastive reasons provide *discrete, condition-like explanations* that directly answer “what is enough to justify this prediction?” or “what must change to flip it?”. Their intuitive nature has given them a central role in many classic XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), shown greater effectiveness in improving human prediction over additive models (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and proved useful for downstream tasks such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A well-established principle further holds that *smaller* sufficient and contrastive explanations enhance interpretability, making *minimality* a central guarantee of interest (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

The complexity of these two explanation types is the backdrop against which our results should be read. Barceló et al. (2020) conducted one of the earliest such studies, establishing that finding the minimal-sized sufficient reason is NP-hard for a decision tree and rises to  $\Sigma_2^P$ -hard for a neural network. For contrastive reasons, they showed that computing the smallest explanation is polynomial-time for decision trees but NP-hard for neural networks. Similar hardness results were later extended to tree ensembles (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b). As we detail below, this hardness persists and even intensifies once one moves to probabilistic and global variants — which is exactly what makes our tractability results for the global setting surprising.

**From non-probabilistic to probabilistic explanations.** The classic definitions above are frequently criticized as too rigid: because they must hold in an absolute sense over entire domains, they can force explanations to be excessively large or uninformative (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To relax this rigidity, the literature has moved to a more general, *probabilistic* notion of sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023), in which one seeks subsets of features that influence a prediction with probability exceeding a threshold  $\delta$ . Our framework is deliberately built at this probabilistic level of generality, so that the classic non-probabilistic case appears simply as the special value  $\delta = 1$ .

This flexibility comes at a steep computational price. Wäldchen et al. (2021) first studied the complexity of probabilistic sufficient reasons, showing the problem is  $\text{NP}^{\text{PP}}$ -hard for CNF classifiers, with the hardness extending to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). The decisive theoretical insight here — one that directly motivates our own analysis — is the *lack of monotonicity* of the probability function (Arenas et al., 2022; Izza et al., 2023), which makes even *subset-minimal* explanations hard to compute. Strikingly, (Arenas et al., 2022) show that finding a subset-minimal probabilistic sufficient reason is NP-hard even for decision trees, in sharp contrast to the non-probabilistic case, where it is polynomial-time (Huang et al., 2021; Barceló et al., 2020). One of our central findings is that this missing monotonicity, and hence tractability, is *restored* in the global setting.

**From local to global explanations.** The final axis we build on is the move from *local* to *global* explanations. Bassan et al. (2024) recently extended the complexity analysis from the *local* (non-probabilistic) setting — where explanations are tied to individual predictions — to the *global* (non-probabilistic) setting, which seeks sufficient or contrastive reasons over entire domains. As with other non-probabilistic methods (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), however, their criteria are extremely strict — arguably even stricter globally than locally. Any feature excluded from a global sufficient reason is deemed strictly redundant (Bassan et al., 2024), so the explanation often spans nearly all input features and becomes uninformative. This is precisely the gap our probabilistic–global framework is designed to close: we retain the appeal of global reasoning while recovering flexibility and, as we show next, tractability.

## OUR CONTRIBUTIONS

This paper makes four contributions, which build on one another: a unifying framework, a structural dichotomy it exposes, and two families of complexity results (one positive, one negative) that follow from that dichotomy.

1. **A unified framework for explanation computation.** We unify previously disparate explanation-computation problems — sufficient and contrastive, probabilistic and non-probabilistic, local and global — into a *single* minimization task over a unified value function. Within this framework, we single out three properties of the value function that dictate the task’s complexity: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity* — three cornerstones of combinatorial optimization.
2. **A local-versus-global structural dichotomy.** Using this framework, we show that these three properties behave in *strikingly different manners* depending on the value function’s structure. Concretely, while the *local* value functions for both sufficient and contrastive reasons are *non-monotonic*, we prove the surprising result that their *global* counterparts are *monotonic non-decreasing*. We further identify structure unique to the global setting: the global sufficient value function is *supermodular* and the global contrastive value function is *submodular* — whereas locally neither property holds. In short, the global value functions possess exactly the structure the local ones lack.
3. **Positive results: tractable global explanations.** We convert this dichotomy into new complexity results, obtaining the intriguing finding that global explanations *with guarantees* can be computed efficiently even though their local counterparts are computationally hard. We establish this across three model types spanning the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. Specifically, we prove that although computing a subset-minimal local probabilistic sufficient/contrastive explanation is NP-hard even for decision trees (Arenas et al., 2022), its *global* counterpart is polynomial-time. We then extend this to *any* black-box model (including neural networks and tree ensembles) under empirical distributions: subset-minimal global sufficient/contrastive explanations are polynomial-time, whereas the local versions remain NP-hard.
4. **Negative results and provable approximations: a hard local frontier.** Finally, we sharpen the picture by exploiting the sub-/super-modularity that holds only globally. We prove that cardinally minimal global explanations admit provable *constant-factor* approximations — even for complex models such as neural networks and tree ensembles — under a fixed empirical distribution. In sharp contrast, we establish strong inapproximability for the local setting: *no* bounded approximation is possible, even under very simple assumptions.

Owing to space constraints, we present the main theorems and corollaries in the body and defer full proofs to the appendix. Throughout, we return to the four contributions above as running signposts, so the reader can track where each result fits.

## 2 PRELIMINARIES

**Setting.** We work over an input space of dimension  $n \in \mathbb{N}$ , with input vectors  $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ ; each coordinate  $\mathbf{x}_i$  ranges over its domain  $\mathcal{X}_i$ , which may be discrete or continuous, so the full feature space is  $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$ . We study classification models  $f : \mathbb{F} \rightarrow [c]$  with  $c \in \mathbb{N}$  classes, together with a *generic* distribution  $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$  over the inputs. Because  $\mathcal{D}$  is often infeasible to approximate exactly, a natural and widely used alternative — one that our tractability results exploit directly — is a fixed empirical dataset  $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$  that serves as a practical proxy for the “true” distribution, e.g., a sample drawn from the training data.

The local-versus-global distinction is central to our contributions, so we fix it early. A *local* explanation targets a specific prediction  $\mathbf{x} \in \mathbb{F}$ , reasoning about why the model predicted  $f(\mathbf{x})$  for that instance. A *global* explanation instead captures the general behavior of  $f$  across a wider region of the input space, independent of any single  $\mathbf{x}$ , characterizing its overall decision-making logic. Our results turn precisely on how the two behave differently.

**Models.** Although many of our results are model-agnostic (e.g., the inherent properties of value functions underlying contribution 2), we also instantiate them into model-specific complexity results

for widely used classifiers. To cover the interpretability spectrum from “black-box” to “interpretable”, we analyze: (i) decision trees, (ii) neural networks (all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, spanning majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). These are exactly the three model families across which our positive results hold. Formal definitions appear in the Appendix.

**Distributions.** In probabilistic explanation settings, complexity hinges on the input distribution  $\mathcal{D}$ , so we distinguish three types that recur throughout our results: (i) *general distributions*, which assume nothing about  $\mathcal{D}$  and thus cover all possibilities — we use these mainly to prove properties that hold universally; (ii) *empirical distributions*, those derived from the finite dataset  $\mathbf{D}$ , an approach standard in XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022) and the setting in which our constant-factor approximations hold; and (iii) *independent distributions*, which assume the features in  $\mathcal{D}$  are mutually independent — another widely adopted XAI assumption (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). Importantly, empirical distributions need not imply feature independence; they can encode complex dependencies extracted from finite datasets (Van den Broeck et al., 2022). Full formal definitions appear in the Appendix.

### 3 FORMS OF REASONS

Before presenting our unified framework (contribution 1), we fix the explanation types it must subsume. We begin with the strict non-probabilistic definitions of (local/global) sufficient and contrastive reasons, and then generalize to their more flexible probabilistic counterparts. Each of the four resulting settings will later be recovered as a single instance of one value-function minimization task.

#### 3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

**Sufficient reasons.** In *feature selection*, users often pick the top features driving a model’s decision. We adopt the well-established *sufficiency* criterion for this choice, consistent with widely used explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022), and we consider it both locally (for a single prediction) and globally (across the entire domain). Following standard conventions,  $S \subseteq \{1, \dots, n\}$  is a *local sufficient reason* when fixing the features in  $S$  to their values in  $\mathbf{x}$  keeps the prediction at  $f(\mathbf{x})$  regardless of the remaining features  $\bar{S}$ . Formally,  $S$  is a local sufficient reason for  $\langle f, \mathbf{x} \rangle$  iff:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here  $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$  denotes the vector taking features in  $S$  from  $\mathbf{x}$  and those in  $\bar{S}$  from  $\mathbf{z}$ . This notion is closely tied to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which seek alternative inputs  $\mathbf{z}'$  that preserve the prediction (i.e.,  $f(\mathbf{z}') = f(\mathbf{x})$ ) while staying as “close” as possible to the original point: instantiating  $\mathbf{z}'$  as  $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$  and measuring proximity via Hamming distance makes the two coincide.

Its global analogue simply demands sufficiency everywhere: a *global sufficient reason* (Bassan et al., 2024) is a subset  $S \subseteq \{1, \dots, n\}$  that is a local sufficient reason for *every* input  $\mathbf{x} \in \mathbb{F}$ :

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

**Contrastive reasons.** The dual approach pinpoints subsets of features that *modify* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024), capturing the minimal change needed to flip the output. Formally,  $S \subseteq \{1, \dots, n\}$  is a *local contrastive reason* for  $\langle f, \mathbf{x} \rangle$  iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Mirroring the sufficient case, contrastive reasons connect to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment  $\mathbf{x}'$  that flips the prediction ( $f(\mathbf{x}') \neq f(\mathbf{x})$ ); under Hamming distance the two notions coincide.

Just as with sufficient reasons, one can ask for a subset that flips the prediction across all inputs in the domain. This global form is closely tied to bias-identification approaches (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and to *group* counterfactual methods that search over multiple instances (Carrizosa et al., 2024; Warren et al., 2024). Formally,  $S$  is a *global contrastive reason* for  $f$  iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

### 3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As noted in the introduction, the non-probabilistic definitions are often criticized as overly strict, motivating a shift to *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021) that require the guarantee to hold only with probability at least  $\delta \in [0, 1]$ . Setting  $\delta = 1$  recovers the original definitions — so, crucially for contribution 1, the probabilistic formulation is the general one that our unified framework operates on.

**Probabilistic sufficient reasons.**  $S \subseteq \{1, \dots, n\}$  is a *local  $\delta$ -sufficient reason* for  $\langle f, \mathbf{x} \rangle$  if, fixing the features in  $S$  to their values in  $\mathbf{x}$  and sampling the rest from  $\mathcal{D}$ , the prediction is preserved with probability at least  $\delta$ :

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where  $\mathbf{z}_S = \mathbf{x}_S$  fixes the features in  $S$  of  $\mathbf{z}$  to their values in  $\mathbf{x}$ . Following the standard notion of global explanations — averaging over the whole domain —  $S \subseteq \{1, \dots, n\}$  is a *global  $\delta$ -sufficient reason* for  $\langle f \rangle$  if the expected local sufficiency probability over  $\mathbf{x} \sim \mathcal{D}$  is at least  $\delta$ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

**Probabilistic contrastive reasons.** Analogously, a *local  $\delta$ -contrastive reason* is a subset that flips the prediction with sufficient probability. Here, unlike the sufficient case, we fix the *complementary* set  $\bar{S}$  to its values in  $\mathbf{x}$  and let the features in  $S$  vary, requiring the prediction to differ from  $f(\mathbf{x})$  with probability exceeding  $\delta$ :

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

As before, the global version averages over the domain:  $S$  is a *global  $\delta$ -contrastive reason* if the expected local contrastive probability over  $\mathbf{x} \sim \mathcal{D}$  exceeds  $\delta$ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

## 4 FORMS OF MINIMALITY

Across every explanation form above — sufficient or contrastive, local or global — the literature holds that *smaller* explanations are more meaningful, making *minimality* a key provable guarantee and the quantity our algorithms optimize. We study two central notions of minimality, both of which apply uniformly to all our explanation types:

**Definition 1.** Assuming a subset  $S \subseteq \{1, \dots, n\}$  is an explanation, then:

1.  $S$  is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff  $S$  has the smallest explanation cardinality  $|S|$  (i.e., there is no explanation  $S'$  such that  $|S'| < |S|$ ).
2.  $S \subseteq \{1, \dots, n\}$  is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff  $S$  is an explanation, and any  $S' \subsetneq S$  is not an explanation.

Cardinal-minimality is strictly stronger than subset-minimality: every cardinally minimal  $S$  is subset-minimal but not conversely (Appendix B.3 gives an example). We say “subset” and “cardinally” minimal, rather than local and global minima, to avoid clashing with the local-vs-global explanation

axis (input- vs. domain-level reasoning). Both notions apply to *every* form we analyze: e.g., a cardinally minimal local probabilistic  $\delta$ -sufficient reason is the one of smallest  $|S|$ , whereas a subset-minimal one is any  $S$  no proper subset of which qualifies. This distinction structures our results: contribution 3 concerns subset-minimality, while contribution 4 concerns the harder cardinal-minimality.

## 5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

We now deliver our first contribution: a single optimization task that subsumes all four explanation settings. The key observation is that every problem discussed so far — local or global, sufficient or contrastive, probabilistic or not — is exactly the search for a minimal-size subset  $S$  with  $v(S) \geq \delta$  (taking  $\delta := 1$  in the non-probabilistic case), for an appropriate value function  $v$ . Fixing notation: let  $v_{\text{suff}}^\ell$  be the local sufficiency probability of equation 5, i.e.,  $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ , with global variant  $v_{\text{suff}}^g$  (equation 6); likewise  $v_{\text{con}}^\ell$  and  $v_{\text{con}}^g$  are the local and global contrastive probabilities (equations 7 and 73). With this single value function  $v$  as the only moving part, the unified task of finding a cardinally minimal  $\delta$ -local/global sufficient/contrastive reason is:

### Cardinally Minimal $\delta$ -Explanation:

**Input:** Model  $f$ , a distribution  $\mathcal{D}$ , (possibly, an input  $\mathbf{x}$ ), a *general* value function  $v : 2^n \rightarrow [0, 1]$  (defined using  $f$ ,  $\mathcal{D}$ , and possibly  $\mathbf{x}$ ), and some  $\delta \in [0, 1]$ .

**Output:** A subset  $S \subseteq [n]$  such that  $v(S) \geq \delta$  and  $|S|$  is minimal.

The same framework covers the relaxed goal of a subset-minimal (rather than cardinally minimal) reason, obtained by swapping only the minimality requirement:

### Subset Minimal $\delta$ -Explanation:

**Input:** Model  $f$ , a distribution  $\mathcal{D}$ , (possibly, an input  $\mathbf{x}$ ), a *general* value function  $v : 2^n \rightarrow [0, 1]$  (defined using  $f$ ,  $\mathcal{D}$ , and possibly  $\mathbf{x}$ ), and some  $\delta \in [0, 1]$ .

**Output:** A subset  $S \subseteq [n]$  such that  $v(S) \geq \delta$  and for any  $S' \subseteq S$  it holds that  $v(S') < \delta$ .

**The three properties of  $v$  that control complexity.** Because every explanation setting is now a single optimization over  $v$ , its complexity is dictated entirely by the structure of  $v$ . We isolate the three properties — announced in contribution 1 — that will determine tractability throughout the paper. The first is *non-decreasing monotonicity*, i.e., the marginal contribution  $v(S \cup \{i\}) - v(S)$  is always non-negative:

**Definition 2.** We say that a value function  $v$  maintains non-decreasing monotonicity if for any  $S \subseteq \{1, \dots, n\}$  and any  $i \in \{1, \dots, n\}$  it holds that:  $v(S \cup \{i\}) \geq v(S)$ .

The remaining two properties, *supermodularity* and *submodularity*, instead concern how that marginal contribution  $v(S \cup \{i\}) - v(S)$  evolves as  $S$  grows: it is non-decreasing in the supermodular case and, dually, non-increasing in the submodular case. Formally:

**Definition 3.** Let there be some value function  $v$ , some  $S \subseteq S' \subseteq \{1, \dots, n\}$ , and  $i \notin S'$ . Then:

1.  $v$  maintains supermodularity iff it holds that:  $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$ .
2.  $v$  maintains submodularity iff it holds that:  $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$ .

## 6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

This section establishes our second contribution — the structural dichotomy between local and global value functions — which underpins all the complexity results that follow. To see why the dichotomy is surprising, recall the local picture. In the non-probabilistic local setting, subset-minimal sufficient or contrastive reasons are computable precisely because monotonicity holds, but only in the restricted case  $\delta = 1$ , which sacrifices practical flexibility. Generalizing to probabilistic guarantees for arbitrary  $\delta$  does not help: prior work shows monotonicity *breaks down* there, making explanations computationally harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b). The remainder of this section shows that the global setting escapes all of these obstructions.

We first clear away a possible objection. In the still more extreme *global non-probabilistic* setting, Bassan et al. (2024) proved a stringent *uniqueness* property — exactly *one* subset-minimal explanation exists — but this again relies on  $\delta = 1$ , which is especially restrictive globally, since the condition must hold for *all* inputs; indeed Bassan et al. (2024) show the unique subset equals the set of all non-redundant features, potentially all of them. Our first structural result shows this uniqueness is an artifact of  $\delta = 1$ : in the general global *probabilistic* setting (any  $\delta$ ) uniqueness *fails*, and the number of subset-minimal explanations can be *exponential*.

**Proposition 1.** *While the non-probabilistic case ( $\delta = 1$ ) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary  $\delta$ ), there exist functions  $f$  that have  $\Theta(\frac{2^n}{\sqrt{n}})$  subset-minimal global sufficient reasons.*

We now state the centerpiece of contribution 2. Although uniqueness is lost, the crucial *monotonicity* property is *gained*: it holds for the global value function at *every*  $\delta$ , for both the sufficient ( $v_{\text{suff}}^g$ ) and contrastive ( $v_{\text{con}}^g$ ) functions. This is exactly the property that fails locally, and its recovery is what drives our tractability results:

**Proposition 2.** *While the local probabilistic setting (for any  $\delta$ ) lacks monotonicity — i.e., the value functions  $v_{\text{con}}^l$  and  $v_{\text{suff}}^l$  are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any  $\delta$ ), both value functions  $v_{\text{con}}^g$  and  $v_{\text{suff}}^g$  are monotonic non-decreasing.*

Monotonicity is only the first of the three properties. Completing contribution 2, we identify the additional structure — submodularity and supermodularity — that appears only globally. For sufficient reasons, under the common feature-independence assumption the global function  $v_{\text{suff}}^g$  is *supermodular*, whereas its local counterpart  $v_{\text{suff}}^l$  is not, even under the far stronger uniform (hence independent) distribution:

**Proposition 3.** *While the local probabilistic sufficient setting (for any  $\delta$ ) lacks supermodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{suff}}^l$  is not supermodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{suff}}^g$  is supermodular.*

The contrastive case completes the symmetry: for global probabilistic *contrastive* reasons the value function is not supermodular but *submodular* — again in sharp contrast to the local setting, where it is neither. Thus each global function carries exactly the modularity property its optimization needs, and the local functions carry none.

**Proposition 4.** *While the local probabilistic contrastive setting (for any  $\delta$ ) lacks submodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{con}}^l$  is not submodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{con}}^g$  is submodular.*

## 7 COMPUTATIONAL COMPLEXITY RESULTS

### 7.1 SUBSET MINIMAL EXPLANATIONS

We now cash in the structural dichotomy of contribution 2 to deliver contribution 3: subset-minimal global explanations are tractable, while their local counterparts are hard. The single property responsible is *monotonicity*, which we proved holds for both global functions  $v_{\text{con}}^g, v_{\text{suff}}^g$  but for neither local one  $v_{\text{con}}^l, v_{\text{suff}}^l$ . This gap is decisive: a simple greedy algorithm provably converges to a subset-minimal explanation in the global setting but can fail locally. The upshot, made precise below across all analyzed model types, is the surprising separation that local subset-minimal explanations are hard whereas global ones are polynomial-time solvable. We start from the following generalized greedy algorithm:

Algorithm 1 searches for a subset-minimal  $\delta$ -explanation top-down. It initializes  $S$  to the full input space  $\{1, \dots, n\}$  and, while some single-feature removal keeps  $v(S \setminus \{i\}) \geq \delta$ , greedily deletes the feature  $j$  that decreases the value function least (i.e., maximizes  $v(S \setminus \{j\})$ ), returning  $S$  at the end.

Whether Algorithm 1 actually returns a subset-minimal explanation again comes down to *monotonicity*. On termination, removing any *single* feature drops  $v(S \setminus \{i\})$  below  $\delta$ ; monotonicity upgrades this to the far stronger guarantee that removing *any* subset  $S'$  does so as well. Combining this with the monotonicity results of contribution 2 yields:

---

**Algorithm 1** Subset Minimal Explanation Search

---

**Input** Value function  $v$ , and some  $\delta \in [0, 1]$ 

```
1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 
```

 $\triangleright S$  is a (subset minimal?)  $\delta$ -explanation

---

**Proposition 5.** *Computing Algorithm 1 with the local value functions  $v_{con}^l$  and  $v_{suff}^l$  does not always converge to a subset minimal  $\delta$ -sufficient/contrastive reason. However, computing it with the global value functions  $v_{con}^g$  or  $v_{suff}^g$  necessarily produces subset minimal  $\delta$ -sufficient/contrastive reasons.*

With this algorithmic tool in hand, we now instantiate contribution 3 as concrete complexity separations for each analyzed model type, beginning with the “interpretable” extreme and building up to complex black-box models.

**Decision trees.** Consider the most favorable, ostensibly “interpretable” case:  $f$  is a decision tree and  $\mathcal{D}$  is independent. Even here a strict local-vs-global separation emerges. Arenas et al. (2022) showed the surprising intractability that, unless  $\text{PTIME}=\text{NP}$ , no polynomial-time algorithm computes a subset-minimal *local*  $\delta$ -sufficient reason for decision trees under independent (even uniform) distributions. Our contribution is the matching positive side: the very same problem is efficiently solvable *globally* — a subset-minimal global  $\delta$ -sufficient reason for decision trees is polynomial-time computable:

**Theorem 1.** *If  $f$  is a decision tree and the probability term  $v_{suff}^g$  can be computed in polynomial time given the distribution  $\mathcal{D}$  (which holds for independent distributions, among others), then obtaining a subset-minimal global  $\delta$ -sufficient reason can be obtained in polynomial time. However, unless  $\text{PTIME}=\text{NP}$ , no polynomial-time algorithm exists for computing a local  $\delta$ -sufficient reason for decision trees even under independent distributions.*

**Extension to other tractable models.** The same global tractability guarantee is not confined to trees: it also holds for *orthogonal DNFs* (Crama & Hammer, 2011) — Disjunctive Normal Form formulas whose terms are pairwise mutually exclusive. This extension matters because orthogonal DNFs *generalize* decision trees while retaining their clean structure, showing our positive results reach beyond tree-structured models. The full argument is in Appendix N.

**Neural networks, tree ensembles, and other complex models.** We now push contribution 3 to its most general form, well beyond decision trees. Under an empirical distribution  $\mathcal{D}$  and the mild assumption that  $f$  admits polynomial-time inference, a subset-minimal global  $\delta$ -sufficient or contrastive reason is computable in polynomial time — for *any* such model:

**Proposition 6.** *For any model  $f$ , and empirical distribution  $\mathcal{D}$  — computing a subset-minimal global  $\delta$ -sufficient or  $\delta$ -contrastive reason for  $f$  can be done in polynomial time.*

Because this positive result holds for *any* model under empirical distributions, it lets us pin down the local-vs-global gap exactly where it is starkest. For complex models, subset-minimal *local* explanations stay intractable even under empirical distributions; establishing this for both neural networks and tree ensembles yields a strict complexity separation, the sharpest statement of contribution 3:

**Theorem 2.** *Assuming  $f$  is a neural network or a tree ensemble, and  $\mathcal{D}$  is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global  $\delta$ -sufficient and contrastive reasons. However, unless  $\text{PTIME}=\text{NP}$ , there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -sufficient reason or a subset minimal local  $\delta$ -contrastive reason.*

## 7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now turn to contribution 4, moving from subset-minimality to the harder *cardinally minimal*  $\delta$ -sufficient/contrastive reason. Here the payoff of the full structural dichotomy becomes clear: globally, the interplay of monotonicity with sub-/super-modularity yields novel *provable approximation*

guarantees, whereas locally the problem is not merely hard but *inapproximable*. This is the strongest form of the global-vs-local separation running through the paper.

**A unified greedy approximation algorithm.** The modularity properties of contribution 2 let us borrow directly from classical optimization. For a monotone non-decreasing *submodular* value function, finding a cardinally minimal explanation is an instance of *submodular set cover* (Wolsey, 1982); for a *supermodular* one it becomes the harder non-submodular variant (Shi et al., 2021). Both are well studied for their strong approximation guarantees (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). We use a single classic greedy algorithm (Algorithm 2) as the common backbone for approximating both cardinally minimal sufficient and contrastive  $\delta$ -reasons, and analyze its guarantees in each case below.

---

**Algorithm 2** Cardinally Minimal Explanation Approximation Search

---

**Input** Value function  $v$ , and some  $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

---

Algorithm 2 mirrors Algorithm 1 but runs bottom-up: starting from the empty set, it greedily adds the feature that increases  $v(S \cup i)$  the least, stopping once the threshold  $\delta$  is reached.

**Cardinally minimal contrastive reasons.** Take global contrastive reasons first, where monotonicity and submodularity both hold. The task is then exactly *submodular set cover*, so Wolsey (1982)’s classic guarantee applies through Algorithm 2: a Harmonic-based factor for integer-valued objectives, and in general an  $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under a fixed-size *empirical* distribution  $\mathcal{D}$  this collapses to a *constant* approximation — only *logarithmic* in the sample size. The local side is the opposite extreme: even for a *single* sample point, *no* bounded approximation exists. This gap is the concrete content of contribution 4 for contrastive reasons.

**Theorem 3.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by  $\mathcal{O}(\ln(|D|))$ , for computing a global cardinally minimal  $\delta$ -contrastive reason for  $f$ , assuming feature independence. In contrast, unless  $\text{PTIME}=\text{NP}$ , no bounded approximation exists for computing a local cardinally minimal  $\delta$ -contrastive reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

We further supply a matching *lower bound* for Theorem 3 in the case of an empirical distribution without any independence assumption (Appendix M), confirming the guarantee is essentially tight.

**Cardinally minimal sufficient reasons.** Sufficient reasons are governed by supermodularity rather than submodularity, and the corresponding supermodular set cover is harder to approximate than its submodular sibling above. Approximation is nonetheless possible when the function has bounded curvature (Shi et al., 2021). Writing the *total curvature* of  $v : 2^n \rightarrow \mathbb{R}$  as  $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$  and invoking Shi et al. (2021), we show Algorithm 2 achieves an  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

As in the contrastive case, a fixed empirical distribution makes this approximation constant. The only price of supermodularity relative to submodularity is the extra  $\frac{1}{1-k^f}$  factor on top of the tighter contrastive  $\mathcal{O}(\ln(|D|))$  bound — the guarantee is still constant-factor. And once more the local variant collapses: it is *inapproximable*, with no *bounded* approximation even when  $|D| = 1$ .

**Theorem 4.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal  $\delta$ -sufficient reason for  $f$ , assuming feature independence. In contrast, unless  $\text{PTIME}=\text{NP}$ , there is no bounded approximation for computing a local cardinally minimal  $\delta$ -sufficient reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

Taken together, contributions 3 and 4 tell one coherent story. Subsection 7.1 showed subset-minimal global explanations are tractable while local ones are hard; this subsection extends the same global-vs-local divide to the harder cardinality minimal task, where global explanations are constant-factor approximable and local ones are inapproximable.

## 8 LIMITATIONS AND FUTURE WORK

A note on the scope of our four contributions is in order. The most general of them — the framework (contribution 1) and the structural properties of value functions (contribution 2) — hold without model or distributional restrictions. The complexity results (contributions 3 and 4) are then instantiated for specific model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions; other settings naturally remain open. We nonetheless view these foundational properties, together with the new tractability and intractability results they yield, as a strong basis for analyzing a broader range of explainability scenarios.

Concretely, our study opens several theoretical questions:

1. *Tightening the positive results (contribution 3).* Can our global tractability guarantees be sharpened for simpler model classes such as decision trees, linear models, or other tractably structured models? Local explanations remain intractable there (Arenas et al., 2022; Subercaseaux et al., 2025), yet the strong global structure we uncovered may permit even better guarantees; tightening the empirical-distribution approximation factors is a further key direction.
2. *Approximating the value function.* When *exact* expectations are infeasible, can techniques such as Fully Polynomial Randomized Approximation Schemes (FPRAS) for DNF formulas (Meel et al., 2019) (hence tree ensembles), or Monte Carlo estimates as in (Subercaseaux et al., 2025), approximate them? The open challenge is preserving our minimality guarantees under approximation error, and identifying the tolerance at which they still hold.
3. *Beyond sufficiency and contrastiveness.* We conjecture that alternative feature-importance notions — e.g., other loss measures such as KL divergence (Conmy et al., 2023) — exhibit the same local-to-global emergence of monotonicity, submodularity, or supermodularity, which would extend our efficient feature-selection results to those metrics as well.

## 9 CONCLUSION

We recap the paper’s four contributions. We introduced a *unified framework* casting all sufficient/contrastive, local/global, probabilistic/non-probabilistic explanation problems as one value-function minimization (contribution 1). Within it, we exposed a stark *structural dichotomy*: the *local* value functions possess none of the combinatorial-optimization properties that make such tasks tractable, yet their *global* counterparts possess all of them — (i) *monotonicity*, (ii) *submodularity* for contrastive reasons, and (iii) *supermodularity* for sufficient reasons (contribution 2). This dichotomy powered a series of surprising *positive results*: global explanations with provable guarantees are efficiently computable, even for complex models such as neural networks (contribution 3). It equally powered matching *negative results*: the corresponding local explanations are NP-hard, and often inapproximable, even in highly simplified scenarios (contribution 4). Together, these results uncover foundational properties of explanations and chart both the tractable and intractable frontiers, opening new avenues for future research.

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# Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

**Appendix A** contains extended related work on Formal XAI.

**Appendix B** contains the formalizations of the models and distributions used in this work.

**Appendix C** contains the proof of Proposition 1.

**Appendix D** contains the proof of Proposition 2.

**Appendix E** contains the proof of Proposition 3.

**Appendix F** contains the proof of Proposition 4.

**Appendix G** contains the proof of Proposition 5.

**Appendix H** contains the proof of Theorem 1.

**Appendix I** contains the proof of Proposition 6.

**Appendix J** contains the proof of Theorem 2.

**Appendix K** contains the proof of Theorem 3.

**Appendix L** contains the proof of Theorem 4.

**Appendix O** contains an LLM usage disclosure.

## A EXTENDED RELATED WORK ON FORMAL XAI

Our work sits within *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), the study of explanations with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025), and specifically within its central theme: the computational complexity of producing such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Because these explanations are typically intractable for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), prior work has largely retreated to more tractable model classes (Marques-Silva & Ignatiev, 2023) — decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026). Against this backdrop, our positive results are notable precisely because they recover tractability for the *global* explanations of even these expressive models.

Beyond complexity theory, a complementary line computes formal explanations in practice with automated reasoning: MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural-network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) for certified explanations of neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Others ease the computational burden via abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), smoothing relaxations (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Closest to our approach is a line that applies combinatorial and submodular optimization to improve computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), chiefly for decision trees. That literature — and our own analysis — observes that the underlying local objective is *not* submodular, so greedy submodular-style methods act only as effective heuristics without guarantees. Our contribution here is to show that the *global* variants are genuinely submodular or supermodular, turning these heuristics into algorithms with formal guarantees and enabling a broader family of principled implementations.

## B MODEL AND DISTRIBUTION FORMALIZATIONS

In this section, we formalize the models and distributions referenced throughout the paper. Specifically, Subsection B.1 defines the model families, while Subsection B.2 formalizes the distributions.

## B.1 MODEL FORMALIZATIONS

In this subsection, we formalize the three base-model types that were analyzed throughout the paper: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

**Decision Trees.** We define a decision tree (DT) as a directed acyclic graph that represents a function  $f : \mathbb{F} \rightarrow [c]$ , where  $c \in \mathbb{N}$  denotes the number of classes. The graph encodes the function as follows: (i) Each internal node  $v$  is assigned a distinct binary input feature from the set  $\{1, \dots, n\}$ ; (ii) Every internal node  $v$  has at most  $k$  outgoing edges, each corresponding to a value in  $[k]$  assigned to  $v$ ; (iii) Along any path  $\alpha$  in the decision tree, each variable appears at most once; (iv) Each leaf node is labeled with a class from  $[c]$ . Thus, assigning values to the inputs  $\mathbf{x} \in \mathbb{F}$  uniquely determines a path  $\alpha$  from the root to a leaf in the DT, where the function output  $f(\mathbf{x})$  corresponds to a class label  $i \in [c]$ . The size of the DT, denoted  $|f|$ , is defined as the total number of edges in the graph. To allow for flexible modeling, the ordering of input variables  $\{1, \dots, n\}$  may differ across distinct paths  $\alpha$  and  $\alpha'$ , ensuring that no variable is repeated along a single path.

**Neural Networks.** We present our *hardness* proofs for neural networks with ReLU activations, though our *tractability results* (i.e., polynomial-time algorithms) apply to any architecture that allows for polynomial-time inference — a standard assumption. Thus, all separation results between tractable and intractable cases we prove carry over to *any neural architecture* at least as expressive as a standard feed-forward ReLU network, encompassing many widely used models such as ResNets, CNNs, Transformers, Diffusion models, and more. Moreover, note that any ReLU network can be represented as a fully connected network by assigning zero weights and biases to missing connections. Following standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025), we thus assume the network is fully connected. Specifically, our analysis applies to multi-layer perceptrons (MLPs). Formally, an MLP  $f$  consists of  $t - 1$  *hidden layers* ( $g^{(j)}$  for  $j = 1, \dots, t - 1$ ) and one output layer  $g^{(t)}$ , where each layer is defined recursively as:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where  $W^{(j)}$  denotes the weight matrix,  $b^{(j)}$  the bias vector, and  $\sigma^{(j)}$  the activation function of the  $j$ -th layer. Accordingly, the model comprises  $t$  weight matrices ( $W^{(1)}, \dots, W^{(t)}$ ),  $t$  bias vectors ( $b^{(1)}, \dots, b^{(t)}$ ), and  $t$  activation functions ( $\sigma^{(1)}, \dots, \sigma^{(t)}$ ).

The input layer is defined as  $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$ , representing the binary input vector. The dimensions of the network are governed by a sequence of positive integers  $d_0, \dots, d_t$ , with weight matrices and bias vectors given by  $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$  and  $b^{(j)} \in \mathbb{Q}^{d_j}$ , respectively. These parameters are learned during training. Since the model functions as a binary classifier over  $n$  features, we set  $d_0 = n$  and  $d_t = 1$ . The hidden layers use the ReLU activation function, defined by  $\text{ReLU}(x) = \max(0, x)$ . Although a sigmoid activation is typically used during training, for interpretability purposes, we assume the output layer applies a threshold-based step function, defined as  $\text{step}(\mathbf{z}) = 1$  if  $\mathbf{z} \geq 0$  and  $\text{step}(\mathbf{z}) = 0$  otherwise.

**Tree Ensembles.** Many popular ensemble methods exist, but since our goal is to provide post-hoc explanations, we focus on the *inference* phase rather than the training process. Our analysis centers on ensemble families that rely on either *majority voting* (e.g., Random Forests) or *weighted voting* (e.g., XGBoost) during inference. However, as with our treatment of neural networks, our *tractability results* — namely, polynomial-time algorithms — extend to any possible ensemble configuration with polynomial-time inference, encompassing an even broader range of ensemble configurations.

**Majority Voting Inference.** In *majority voting* inference, the final prediction  $f(\mathbf{x})$  is assigned to the class  $j \in [c]$  that receives the majority of votes among the individual predictions  $f_i(\mathbf{x})$  from all  $i \in [k]$  (i.e., from each tree in the ensemble).

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where  $\text{majority}(S)$  denotes the most frequent label in the multiset  $S$ . If there is a tie, it is resolved by a fixed tie-breaking rule (e.g., lexicographic order or predefined priority).

**Weighted Voting Inference.** In *weighted voting* inference, each model in the ensemble is assigned a weight  $\phi_i \in \mathbb{Q}$  representing its relative importance. The predicted class is the one with the highest total weight across all models. Formally, for any  $\mathbf{x} \in \mathbb{F}$ , we define  $f$  as:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where  $\mathbb{I}$  denotes the identity function.

## B.2 DISTRIBUTION FORMALIZATIONS

This subsection formalizes the distribution definitions discussed in the main paper.

**Empirical Distributions.** The distribution  $\mathcal{D}$  over the input features will be defined based on a dataset  $\mathbf{D}$  comprising various inputs  $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$ . Here, the distribution of any given input  $\mathbf{x} \in \mathbb{F}$  is defined by the frequency of occurrence of  $\mathbf{x}$  within  $\mathbf{D}$ , specifically by:

$$\mathbf{Pr}(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

**Independent Distributions.** Formally, given a probability value  $p(\mathbf{x}_i) \in [0, 1]$  defined for each individual input feature, we say that the distribution  $\mathcal{D}$  is independent if the joint probability over inputs is given by  $\mathbf{Pr}(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$ .

We observe that when  $p(\mathbf{x}_i) = p(\mathbf{x}_j)$  holds for all  $i, j \in [n]$ , the distribution reduces to the *uniform distribution*, a special case of independent distributions.

**General Distributions.** We note that many of the proofs in this work — particularly those concerning fundamental properties of value functions — apply broadly over general distribution assumptions. While empirical distributions (using the training dataset as a proxy) are common in XAI, alternative frameworks for approximating distributions include k-NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for modeling tabular dependencies (Größer & Okhrin, 2022), or more advanced conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022). Such choices are common in the counterfactual explanation and algorithmic recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which are related to contrastive explanations. Furthermore, when the value function is not computed directly from a structural property of the model (e.g., leaf enumeration in decision trees) or from empirical distributions, one may instead approximate it using methods such as Monte Carlo sampling (Hastings, 1970), following approaches similar to (Subercaseaux et al., 2025; Lopardo et al., 2023).

## B.3 SUBSET VS. CARDINAL MINIMALITY

We elaborate here on the distinction between cardinal and subset minimality, which separates contribution 4 (cardinal) from contribution 3 (subset). Cardinal minimality is the substantially stronger guarantee, corresponding to a globally minimal explanation size, whereas subset minimality guarantees only a local form of minimality.

To see why cardinal minimality is stronger, note that if  $S \subseteq [n]$  is a cardinally minimal explanation, then no smaller set  $S'$  with  $|S'| < |S|$  can qualify as an explanation. Hence, no strict subset  $S' \subsetneq S$  is an explanation, which means  $S$  is also subset minimal.

However, the reverse does not hold. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

and the assignment  $\mathbf{x} := (1, 1, 1)$ , which gives  $f(1, 1, 1) = 1$ . Fixing feature  $\mathbf{x}_1 = 1$  yields a cardinally minimal (and subset minimal) sufficient reason, since the prediction remains 1 regardless

of  $\mathbf{x}_2, \mathbf{x}_3$ . But fixing both  $\mathbf{x}_2 = 1$  and  $\mathbf{x}_3 = 1$  also gives a subset minimal explanation—yet not a cardinally minimal one. Thus, while every cardinally minimal explanation is subset minimal, not every subset minimal explanation is cardinally minimal.

## C PROOF OF PROPOSITION 1

**Proposition 1.** *While the non-probabilistic case ( $\delta = 1$ ) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary  $\delta$ ), there exist functions  $f$  that have  $\Theta(\frac{2^n}{\sqrt{n}})$  subset-minimal global sufficient reasons.*

*Proof.* It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input  $\mathbf{x}$  Bassan et al. (2024). In particular, this phenomenon was shown to occur in functions of the following form. We demonstrate that the same function admits an exponential number of *global* and *probabilistic* sufficient reasons. Notably, the mentioned function is a special case of a broader class of *threshold Boolean functions* described in Wegener (2005). Specifically, for  $n = 2k + 1$  with  $k \in \mathbb{N}$ , this function is defined as:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where  $\mathbf{x}$  is drawn from a uniform distribution  $\mathcal{D}$ . Notably, the function  $f$  is *symmetric*, in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends solely on the number of 1’s (or, equivalently, 0’s) in the input and is invariant under any permutation of input bits. Furthermore, the two extreme inputs — the all-zeros vector  $(0, \dots, 0)$  and the all-ones vector  $(1, \dots, 1)$  — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Since the function  $f$  is symmetric, the condition  $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$  holds for any two subsets  $S_1, S_2 \subseteq [n]$ . Moreover, it can be verified that for any subset  $S \subseteq [n]$  of size  $k + 1$ , we have:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set  $\delta$  to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and also set  $k := \lfloor \frac{n}{2} \rfloor$ . Each of these subsets is *subset-minimal* because any subset of size at most  $\lfloor \frac{n}{2} \rfloor - 1$  fails to be a sufficient reason for  $\langle f, \mathbf{x} \rangle$  (with respect to the  $\delta$  threshold). Therefore, we can directly apply the same analysis as in (Bassan et al., 2024). In particular, there are exactly  $\binom{n}{\lfloor \frac{n}{2} \rfloor}$  subset-minimal local sufficient reasons for  $\langle f, \mathbf{x} \rangle$ . Using Stirling’s approximation, we obtain:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This yields the corresponding bound on the number of subset-minimal global  $\delta$ -sufficient reasons.

□

## D PROOF OF PROPOSITION 2

**Proposition 2.** *While the local probabilistic setting (for any  $\delta$ ) lacks monotonicity — i.e., the value functions  $v_{\text{con}}^\ell$  and  $v_{\text{suff}}^\ell$  are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any  $\delta$ ), both value functions  $v_{\text{con}}^g$  and  $v_{\text{suff}}^g$  are monotonic non-decreasing.*

*Proof.* In this section, we prove the monotonicity property for both types of global explanations — sufficient and contrastive. We begin by establishing the property for global sufficiency. To build

intuition, we first focus on the simpler case of Boolean functions, then generalize the result to functions with discrete multi-valued input domains and multiple output classes. We further extend the result to continuous input domains and show that the monotonicity property holds for any well-defined classification function. Finally, we demonstrate how the same monotonicity guarantees apply to global contrastive explanations. Importantly, none of our proofs rely on the assumption of feature independence — the monotonicity property holds strongly for *any* underlying distribution.

Lastly, we present a simple example showing that the monotonicity property does not hold for the local probabilistic sufficient or contrastive value functions.

**Lemma 1.** *The global sufficient value function  $v_{\text{suff}}^g$  is monotonic non-decreasing for Boolean functions, under any data distribution.*

*Proof.* Given an arbitrary set  $S \in [n]$ , and a feature  $i \notin S$ , we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points  $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$  such that  $\mathbf{x}_S = \mathbf{x}'_S$  and  $f(\mathbf{x}) = 1 - f(\mathbf{x}')$ , it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

To simplify notation, we occasionally omit the distribution notation  $\sim \mathcal{D}$  in the local probability expression  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$ , and use  $f^+$  and  $f^-$  to denote the events  $f(\mathbf{z}) = 1$  and  $f(\mathbf{z}) = 0$ , respectively.

We begin with a technical simplification of the probability term to facilitate the proof, and then proceed to establish monotonicity.

**Simplifying the probability term.** Let  $\mathcal{D}_S$  denote the distribution restricted to the set  $S$ . Given a point  $\mathbf{x} \sim \mathcal{D}$  and a set  $S$ , consider all points  $\mathbf{z} \sim \mathcal{D}$  such that  $\mathbf{z}_S = \mathbf{x}_S$  and  $f(\mathbf{z}) = b$ , where  $b \in \{0, 1\}$  — they all share the same local probability  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$ . For every  $\mathbf{x} \sim \mathcal{D}$  that shares the same  $x_S$  and the same output  $b$ , we marginalize over  $x_{\bar{S}}$ . This yields  $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$ , from which we can infer:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ , we get that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2], \end{aligned} \quad (22)$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

**Proving monotonicity.** We now proceed to prove the monotonicity claim, building on the previous simplification. We begin by introducing a few additional notations. Let  $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$  and  $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$ . Similarly, define  $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$  and  $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$ . Then, the following expectations  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$  and  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$  can be simplified accordingly<sup>1</sup>:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

Let  $\Delta_t(S, f)$  denote  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ . Without loss of generality, we assume that both  $P_1$  and  $P_0$  are positive. Otherwise, it can be verified that  $\Delta_t(S, f) = 0$ . Then, we get:

<sup>1</sup>Note that the notations  $P_1$ ,  $P_0$ ,  $g_1^+$ , and  $g_0^+$ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input  $\mathbf{x}_S$ . Formally, they should be written as  $P_1(\mathbf{x}_S)$ ,  $P_0(\mathbf{x}_S)$ , and so on. However, for readability, we omit the explicit dependence on  $\mathbf{x}_S$ .

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\
&\geq 0.
\end{aligned} \tag{26}$$

Note that  $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$  and similarly:  $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$ . Using these identities along with the previous result, we conclude that  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$  is monotone, thereby concluding our proof.  $\square$

**Lemma 2.** *The global sufficient value function  $v_{\text{uff}}^g$  is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

*Proof.* Extending the proof from the simplified Boolean case, we now generalize the result to functions whose input domains and output ranges each consist of a finite set of values. Let  $\mathcal{K}$  represent the set of output classes.

We can generalize equation 22 as follows:

$$\begin{aligned}
\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] &= \\
\sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right]
\end{aligned} \tag{27}$$

Given the condition  $\mathbf{z}_S = \mathbf{x}_S$ , suppose feature  $t$  can take  $r$  possible values  $\{v_1, \dots, v_r\}$ . Then, equation 23 can be generalized as follows:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{k=1}^r \left[ \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_t = v_k \mid \mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right].
\end{aligned} \tag{28}$$

Let  $P_k$  denote  $\Pr(z_t = v_k \mid \mathbf{x}_S)$ , Let  $g_k^j$  denote  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$ , that is,  $z_t$  takes the value  $v_k$  and the output class is  $j$ . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{k=1}^r P_k \cdot \left( \sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
1296 \quad & \Delta_t(S, f) \\
1297 \quad & = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{k=1}^r P_k \cdot \left( \sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
1298 \quad & = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \quad (30) \\
1299 \quad & = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
1300 \quad & = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
1301 \quad & \geq 0.
\end{aligned}$$

Hence,  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$  is monotone, which completes the proof for this setting.  $\square$

**Lemma 3.** *The global sufficient value function  $v_{\text{uff}}^g$  is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

*Proof.* We extend the monotonicity results to functions where features are defined over continuous domains. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in lemma 2 by revising the notation accordingly. The core reasoning remains unchanged when transitioning from the discrete to the continuous case—only the notation differs.

Alternatively, we provide a direct proof below. Assume all features lie in continuous domains, and let  $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$  denote the corresponding joint probability density function. Moreover, let  $p(x_i)$  denote the marginal probability density function of  $x_i$ , and  $\mathcal{X}_i$  denote the domain of feature  $i$ .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \quad (31)$$

where  $\mathcal{X}_{\bar{S}}$  represents the domain of features in  $\bar{S}$ .

And the conditional density is defined as:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \quad (32)$$

The local probability of  $S$  given  $\mathbf{x}$  is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \quad (33)$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \quad (34)$$

via an additional feature  $t \notin S$ .

We then modify equation 27 to

$$\begin{aligned}
1347 \quad & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
1348 \quad & = \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S, \quad (35) \\
1349 \quad &
\end{aligned}$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left( \sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let  $Y_j(t)$  denote  $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$ , and  $p_{t|S}$  denote  $p(z_t \mid \mathbf{x}_S)$ . Note that both  $Y_j(t)$  and  $p_{t|S}$  are associated with a fixed  $\mathbf{x}_S$ <sup>2</sup>. We prove monotonicity as follows:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left( \sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left( \int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

**Proposition 2.** While the local probabilistic setting (for any  $\delta$ ) lacks monotonicity — i.e., the value functions  $v_{\text{con}}^\ell$  and  $v_{\text{suff}}^\ell$  are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any  $\delta$ ), both value functions  $v_{\text{con}}^g$  and  $v_{\text{suff}}^g$  are monotonic non-decreasing.

*Proof.* As a consequence of lemmas 1 to 3, we conclude that the global sufficient value function  $v_{\text{suff}}^g$  is monotonic non-decreasing for any well-defined classification function — whether its features are Boolean, discrete, continuous, or a combination — under any data distribution.

$$v_{\text{con}}^g(S) = 1 - v_{\text{suff}}^g(\bar{S}) \quad (38)$$

This implies that  $v_{\text{con}}^g$  is monotonic non-increasing with respect to adding features to  $\bar{S}$ . Equivalently,  $v_{\text{con}}^g$  is monotonic non-decreasing as features are removed from  $\bar{S}$  and added to  $S$ , thus completing the proof.

□

## E PROOF OF PROPOSITION 3

**Proposition 3.** While the local probabilistic sufficient setting (for any  $\delta$ ) lacks supermodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{suff}}^\ell$  is not supermodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{suff}}^g$  is supermodular.

*Proof.* This section establishes the supermodularity property for both global sufficient and contrastive explanations. As with monotonicity, we begin with Boolean functions and progressively generalize

<sup>2</sup>Changing  $\mathbf{x}_S$  alters their values, we omit  $\mathbf{x}_S$  in the notation for brevity.

to multi-valued discrete and continuous input domains, and finally consider the case of well-defined classification functions. Unlike the monotonicity property, however, these proofs require the assumption of feature independence. We also provide a counterexample to show that supermodularity may fail when this assumption is relaxed.

**Lemma 4.** *The global sufficient value function  $v_{\text{uff}}^g$  is supermodular for Boolean functions, under the assumption of feature independence.*

*Proof.* Particularly, we will prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

where  $S \subseteq S'$  and  $t \notin S'$ . That is,  $\Delta_t(S, f)$  is monotone.

Clearly, if  $S' = S$ , then  $\Delta_t(S, f) \leq \Delta_t(S', f)$  holds. Let  $S' = S \cup \{i\}$ , where  $i \notin S$  and  $i \neq t$ . For the local probability  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$ , under the assumption of feature independence, we have that the following holds:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \\ = \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let  $P_1, P_0, Q_1$ , and  $Q_0$  denote  $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$ , and  $\Pr(z_i = 0)$ , respectively. Moreover, let  $g_{11}^+, g_{10}^+, g_{01}^+$ , and  $g_{00}^+$  denote  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$ , and  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$ , respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This implies that  $\Delta_t(S, f)$  is monotone, that is,  $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$ .  $\square$

**Lemma 5.** *The global sufficient value function  $v_{\text{uff}}^g$  is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

*Proof.* We extend the results of the previous section to multi-valued discrete functions, again with the assumption of feature independence. We use  $P_k$  to denote  $\Pr(z_t = v_k)$ , and  $Q_{k'}$  to denote  $\Pr(z_i = v_{k'})$ . Moreover, we use  $g_{k'k}^j$  to denote  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}, z_t = v_k)$ . We

decompose equation 30 to

$$\begin{aligned}\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k P_l \cdot \left( \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k P_l \cdot \left( \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right].\end{aligned}\quad (44)$$

Likewise, we have

$$\begin{aligned}\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[ \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_i = v_k) \cdot \sum_{j \in \mathcal{K}} \left( \sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left( \sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].\end{aligned}\quad (45)$$

Therefore, we get

$$\begin{aligned}\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\ &\geq 0.\end{aligned}\quad (46)$$

□

**Proposition 3.** *While the local probabilistic sufficient setting (for any  $\delta$ ) lacks supermodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{diff}}^\ell$  is not supermodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{diff}}^g$  is supermodular.*

*Proof.* We can generalize the supermodularity results established in Lemma 5 to the continuous case, assuming that  $\mathcal{D}_p$  satisfies feature independence. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in Lemma 5 by revising the notation accordingly.

Alternatively, we provide a direct proof below. Under this assumption, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \quad (47)$$

Consequently, the local probability of  $S$  conditioned on  $\mathbf{x}$  becomes:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \quad (48)$$

We let  $Y_j(i, t)$  denote  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$ , where  $i \notin S, t \notin S$  and  $i \neq t$ . Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
\Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left( \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left( \int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left( \int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\
&= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}.
\end{aligned}$$

Let  $S' = S \cup \{i\}$ , then we have:

$$\begin{aligned}
\Delta_t(S', f) &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \tag{50} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S,
\end{aligned}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\
&= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}.
\end{aligned}$$

Let  $\alpha(i, t, t')$  denote  $Y_j(i, t) - Y_j(i, t')$  and  $\beta(i', t, t')$  denote  $Y_j(i', t) - Y_j(i', t')$ , we can infer that

$$\begin{aligned}
\Delta_t(S', f) - \Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[ \sum_{j \in \mathcal{K}} \frac{1}{4} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&\geq 0.
\end{aligned} \tag{51}$$

Consequently, the global sufficient value function  $v_{\text{suff}}^g$  retains the supermodularity property for any valid classification function, regardless of whether its features are Boolean, discrete, continuous, or a mix thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables  $\{x_1, x_2, x_3\}$  and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \tag{52}$$

Consider the input  $\mathbf{x} = (1, 1, 1)$  which is classified as 1. For the local probabilistic sufficient value function  $v_{\text{suff}}^l$ , we have that the following holds:

$$v_{\text{suff}}^l(\emptyset) = \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4},$$

$$v_{\text{suff}}^l(\{1, 2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1.$$

Since  $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$ , the function  $v_{\text{suff}}^l$  is neither supermodular nor submodular.

□

## F PROOF OF PROPOSITION 4

**Proposition 4.** *While the local probabilistic contrastive setting (for any  $\delta$ ) lacks submodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{con}}^l$  is not submodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{con}}^g$  is submodular.*

*Proof.* Once we establish that the global sufficient setting is supermodular, it directly follows that the global contrastive setting is submodular. Let  $S \subseteq [n]$  be the set of features allowed to change, and suppose  $S \subseteq S'$  with  $i \notin S'$ . To compare the marginal contributions  $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$  and  $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$ , we equivalently compare  $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$  and  $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$ . Since  $S \subseteq S'$ , it follows that  $\bar{S}' \subseteq \bar{S}$ , and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function  $v_{\text{con}}^g$  is submodular, hence concluding this proof.

We will now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables  $\{x_1, x_2, x_3\}$  and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input  $\mathbf{x} = (1, 1, 1)$  which is classified as 1. For the local probabilistic contrastive value function  $v_{\text{con}}^l$ , we have:

$$v_{\text{con}}^l(\emptyset) = 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2},$$

$$v_{\text{con}}^l(\{1, 2\}) = \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}.$$

Since  $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$ , the function  $v_{\text{con}}^l$  is neither submodular nor supermodular.

□

In the rest of this section, we show that the feature independence assumption is not just sufficient but also necessary for supermodularity. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this boolean function, it holds that  $f(00) = 0$ ,  $f(01) = 0$ ,  $f(10) = 1$ , and  $f(11) = 1$ . Moreover, define a distribution  $\mathcal{D}$  on  $x_1$  and  $x_2$ . Recall that the global sufficient value function is:

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$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where  $S \subseteq S'$  and  $t \notin S'$ , then the global sufficient value function is supermodular. When  $S = \emptyset$ , we have that the following holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) \\ &= (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \end{aligned}$$

When  $S = \{2\}$ , we have that the following condition holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1) \end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\ &= \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \end{aligned}$$

In contrast, when  $S = \{1\}$ , we have the the following holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) \end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when  $S = \{1, 2\}$ , we have the following condition:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0)} + \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 1, x_2 = 0)} + \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 1)} \\
& = 1
\end{aligned}$$

Let  $t = 2$ . Evidently, we have  $\Delta_t(\{1\}, f) = 0$ . Next, let us analyse  $\Delta_t(\emptyset, f)$ . We note that if we assume feature independence, we observe that:

$$\begin{aligned}
& (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2 \\
& = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned}
& \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2.
\end{aligned}$$

This implies  $\Delta_t(\emptyset, f) = 0$ , that is,  $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$ . However, without the assumption of feature independence, we cannot guarantee  $\Delta_t(\emptyset, f) = 0$ . For example, let:

$$\Pr(x_1 = 0, x_2 = 0) = P_{00}, \Pr(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\Pr(x_1 = 1, x_2 = 0) = P_{10}, \Pr(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying  $\Delta_t(\emptyset, f)$  we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[ \frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[ (P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let  $P_{00} = 0.4$ ,  $P_{01} = 0.2$ ,  $P_{10} = 0.1$ , and  $P_{11} = 0.3$ , and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[ \frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[ (0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that  $\Delta_t(\emptyset, f) > \Delta_t(1, f)$ , indicating that — unlike in the monotonicity setting where it was not required — feature independence is a necessary condition for supermodularity.

## G PROOF OF PROPOSITION 5

**Proposition 5.** *Computing Algorithm 1 with the local value functions  $v_{\text{con}}^\ell$  and  $v_{\text{suff}}^\ell$  does not always converge to a subset minimal  $\delta$ -sufficient/contrastive reason. However, computing it with the global value functions  $v_{\text{con}}^g$  or  $v_{\text{suff}}^g$  necessarily produces subset minimal  $\delta$ -sufficient/contrastive reasons.*

*Proof.* We note that this proposition follows directly from our proof that both  $v_{\text{suff}}^g$  and  $v_{\text{con}}^g$  are monotone non-decreasing (as established in Proposition 2), along with other relevant conclusions drawn in prior work. In the remainder of this section, we elaborate on how this monotonicity property leads to the stated corollary.

The convergence of Algorithm 1 to a subset-minimal  $\delta$  global sufficient or contrastive reason with respect to the value functions  $v_{\text{con}}^g$  and  $v_{\text{suff}}^g$  follows directly from the monotonicity property, for which prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) showed that this type of greedy algorithm yields a subset-minimal explanation when the underlying value function is monotone non-decreasing. The algorithm halts when, for all  $i \in S$ , the value  $v(S \setminus \{i\})$  drops below  $\delta$ , ensuring that although  $v(S) \geq \delta$  (a maintained invariant), removing any element causes the condition to fail. Due to monotonicity, this implies that no proper subset of  $S$  satisfies the condition, guaranteeing that  $S$  is subset-minimal.

We now turn to explain why Algorithm 1 does not converge to a subset-minimal solution in the local setting. We note that it is well known that the local probabilistic sufficient value function (and likewise the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution on the two binary variables  $\{x_1, x_2\}$  and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input  $\mathbf{x} = (0, 1)$  which is classified as 1, then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Because  $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$ , the function  $v_{\text{suff}}^\ell$  is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For  $v_{\text{con}}^\ell$ , the  $S$  denotes the set of features that are allowed to vary.) Because  $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$ , the function  $v_{\text{con}}^\ell$  is also non-monotone.

□

## H PROOF OF THEOREM 1

**Theorem 1.** *If  $f$  is a decision tree and the probability term  $v_{\text{suff}}^g$  can be computed in polynomial time given the distribution  $\mathcal{D}$  (which holds for independent distributions, among others), then obtaining a subset-minimal global  $\delta$ -sufficient reason can be obtained in polynomial time. However, unless  $\text{PTIME}=\text{NP}$ , no polynomial-time algorithm exists for computing a local  $\delta$ -sufficient reason for decision trees even under independent distributions.*

*Proof.* We begin by noting that for decision trees, assuming that  $\mathcal{D}$  represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

Can be carried out in polynomial time using the following procedure. Thanks to the tractability of decision trees, we can iterate over all pairs of leaf nodes, each corresponding to partial assignments

$\mathbf{x}_S$  and  $\mathbf{z}_{S'}$ . For each such pair, if both leaf nodes yield the same prediction under  $f$ , we compute the corresponding term in the expected value  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$  by multiplying the respective feature-wise probabilities over the shared features of the two vectors. Under the feature independence assumption, these probabilities decompose, allowing the full expectation to be computed by summing the contributions of all such matching leaf pairs.

Since all probabilities involved are provided as part of the input and each step involves only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step in Algorithm 1 is efficient, and thus the algorithm as a whole runs in polynomial time. Finally, combining this with Lemma 2, which proves that  $v_{\text{suff}}^g$  is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation. This is because the algorithm halts when, for all  $i \in S$ , the value  $v(S \setminus \{i\})$  falls below  $\delta$ , while  $v(S) \geq \delta$  is preserved throughout. By monotonicity, this guarantees that no strict subset of  $S$  satisfies the condition, ensuring  $S$  is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which showed that, assuming  $P \neq NP$ , there is no polynomial-time algorithm for computing a subset-minimal  $\delta$ -sufficient reason for decision trees. While this was proven specifically under the uniform distribution, the hardness clearly extends to independent distributions, which include this case. Combined, these results establish the complexity separation between the local and global variants, thus concluding our proof.

□

## I PROOF OF PROPOSITION 6

**Proposition 6.** *For any model  $f$ , and empirical distribution  $\mathcal{D}$  — computing a subset-minimal global  $\delta$ -sufficient or  $\delta$ -contrastive reason for  $f$  can be done in polynomial time.*

*Proof.* The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

Can be performed in polynomial time when  $\mathcal{D}$  is selected from the class of empirical distributions. This is achieved by iterating over pairs of instances  $\mathbf{x}, \mathbf{z}$  within the dataset and running an inference through  $f(\mathbf{x})$  and  $f(\mathbf{z})$  to compute the expected values at each step. Consequently, determining whether the probability function exceeds or falls below a given threshold  $\delta$  can also be accomplished in polynomial time. Furthermore, leveraging the proof of non-decreasing monotonicity (Proposition D) of the global probability function, we can utilize algorithm 1 for contrastive reason search or for sufficient reason search. By executing a linear number of polynomial queries, a subset-minimal explanation (whether a subset minimal contrastive reason or a subset minimal sufficient reason) can thus be obtained in polynomial time.

□

## J PROOF OF THEOREM 2

**Theorem 2.** *Assuming  $f$  is a neural network or a tree ensemble, and  $\mathcal{D}$  is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global  $\delta$ -sufficient and contrastive reasons. However, unless  $P = NP$ , there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -sufficient reason or a subset minimal local  $\delta$ -contrastive reason.*

*Proof.* The first part of the proof follows directly from Proposition 6, which established that this holds for any model, since both  $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$  and  $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$  can be computed in polynomial time from

the empirical distribution. For the second part, we proceed by proving two lemmas: one establishing the intractability of computing the sufficient case, and the other addressing the contrastive case.

**Lemma 6.** *Unless  $P_{TIME} = NP$ , then there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

*Proof.* We will first prove this claim for neural networks and then extend the result to tree ensembles. We will establish this claim by demonstrating that if a polynomial-time algorithm exists for computing a subset-minimal local  $\delta$  sufficient reason for neural networks, then it would enable us to solve the classic NP-complete *CNF-SAT* problem in polynomial time. The *CNF-SAT* problem is defined as follows:

**CNF-SAT:**

**Input:** A formula in conjunctive normal form (CNF):  $\phi$ .

**Output:** Yes, if there exists an assignment to the  $n$  literals of  $\phi$  such that  $\phi$  is evaluated to True, and No otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

**Lemma 7.** *Any boolean circuit  $\phi$  can be encoded into an equivalent MLP over the binary domain  $\{0, 1\}^n \rightarrow \{0, 1\}$  in polynomial time.*

*Proof.* We will actually establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution  $\mathcal{D}$  consists of a single element, which, for simplicity, we take to be the zero vector  $\mathbf{0}_n$ . Since the sufficiency of  $S$  in this scenario depends only on the local instance  $\mathbf{x}$  and the single instance  $\mathbf{0}_n$  in our empirical distribution, the nature of the input format (whether discrete, continuous, etc.) does not affect the result. Therefore, the hardness results hold universally across all these settings.

Similarly to  $\mathbf{0}_n$ , we define  $\mathbf{1}_n$  as an  $n$ -dimensional vector consisting entirely of ones. Given an input  $\phi$ , we initially assign  $\mathbf{1}_n$  to  $\phi$ , effectively setting all variables to True. If  $\phi$  evaluates to True under this assignment, then a satisfying truth assignment exists. Therefore, we assume that  $\phi(\mathbf{1}_n) = 0$ . We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

$\phi'$ , while no longer a CNF, can still be transformed into an MLP  $f$  using Lemma 8, ensuring that  $f$  behaves equivalently to  $\phi'$  over the domain  $\{0, 1\}^n$ . Given our assumption that computing a subset-minimal local  $\delta$ -sufficient reason is feasible in polynomial time, we can determine one for the instance  $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$ ,  $\delta := 1$ , noting that the empirical distribution  $\mathcal{D}$  is simply defined over the single data point  $\mathbf{0}_n$ .

We now assert that the subset-minimal  $\delta$ -sufficient reason generated for  $\langle f, \mathbf{x} \rangle$  encompasses the entire input space, i.e.,  $S = \{1, \dots, n\}$ , if and only if  $\langle \phi \rangle \notin \text{CNF-SAT}$ .

Let us assume that  $S = \{1, \dots, n\}$ . Since  $S$  is a  $\delta$ -sufficient reason for  $\langle f, \mathbf{x} \rangle$ , this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is  $\emptyset$  in this case, this trivially holds. The fact that  $S$  is subset-minimal means that any other subset  $S' \subseteq S$  satisfies  $v(S') < \delta = 1$ . Since the probability function  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{x}_S = \mathbf{z}_S)$  is determined by a single point (the distribution contains only the point  $\mathbf{0}_n$ ), the probability function can only take values of 1 or 0. Hence, we also know that  $v(S') = 0$ . This tells us that, aside from the subset  $S = \{1, \dots, n\}$ , for any subset  $S' \subseteq S$ , fixing the features in  $S'$  to 1 and the rest to 0 does not result in a classification outcome of 1. Since the  $\phi_2$  component within  $\phi'$  is True only if all features are assigned 1, this directly implies that  $\phi$  is assigned False for any of these inputs. Since we already know that  $\phi$  does not return a True answer for the vector assignment  $\mathbf{1}_n$  (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that  $\phi \notin \text{CNF-SAT}$ .

Now, suppose that  $S$  is a subset that is strictly contained within  $\{1, \dots, n\}$ . Given that  $S$  is sufficient under our definitions of the distribution  $\mathcal{D}$  with  $\delta = 1$ , we can apply the same reasoning as before to

conclude that  $v(S) = 1$ . This implies that setting the features in  $S$  to 1 while setting the remaining features to 0 ensures that the function  $f$  evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the  $\phi_2$  component within  $\phi'$  must be False. Consequently, the  $\phi$  component must be True, which implies that  $\langle \phi \rangle \in \text{CNF-SAT}$ . This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

**Lemma 8.** *Any CNF or DNF  $\phi$  can be encoded into an equivalent random forest classifier over the binary domain  $\{0, 1\}^n \rightarrow \{0, 1\}$  in polynomial time.*

*Proof.* We observe that we can apply the same process used in our proof for neural networks, where we encoded  $\phi'$  into an equivalent neural network. However,  $\phi'$  is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since  $\phi'$  consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

Where each  $c_i$  is a disjunction of a few terms. Consequently,  $\phi'$  is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

**Lemma 9.** *Unless  $\text{PTIME} = \text{NP}$ , then there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

*Proof.* We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula  $\phi$ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of  $\phi$ .

First, we check whether assigning all variables in  $\phi$  to 1 evaluates  $\phi$  to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network  $f$ . Next, we compute a subset-minimal  $\delta$ -sufficient reason by setting  $\mathcal{D}$  as the empirical distribution containing only a single data point  $\mathbf{0}_n$ , following a similar procedure to Lemma 6. Additionally, we set  $\delta = 1$  and compute a subset-minimal  $\delta$ -contrastive reason concerning  $\langle f, \mathbf{x} \rangle$ .

We will now demonstrate that if any subset-minimal  $\delta$  contrastive reason obtained for  $\phi$  is satisfiable, then  $\phi$  itself is satisfiable. Conversely, if no subset-minimal  $\delta$  contrastive reason is obtained, then  $\phi$  is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ , where the distribution  $\mathcal{D}$  considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a  $\delta = 1$  contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of  $f$ , which corresponds to modifying the assignment of  $\phi$  from False to True. If such an assignment exists, then  $\phi$  is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates  $f$  (and consequently  $\phi$ ) to True, implying that  $\phi$  is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

## K PROOF OF THEOREM 3

**Theorem 3.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by  $\mathcal{O}(\ln(|D|))$ , for computing a global cardinally minimal  $\delta$ -contrastive reason for  $f$ , assuming feature independence. In contrast, unless  $\text{PTIME}=\text{NP}$ , no bounded approximation exists for computing a local cardinally minimal  $\delta$ -contrastive reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

*Proof.* We divide the proof of the theorem into two lemmas, covering both the approximation guarantee and the result on the absence of a bounded approximation.

**Lemma 10.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by  $\mathcal{O}(\ln(|D|))$ , for computing a global cardinally minimal  $\delta$ -contrastive reason for  $f$ , assuming feature independence.*

*Proof.* We will in fact prove a stronger claim, showing that this holds for *any* model, provided the trivial condition that its inference time is computable in polynomial time, along with one additional mild condition that we will detail later — both of which apply to both our neural network and tree ensemble formalizations.

We begin by noting that, since we are working with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs  $\mathbf{x}, \mathbf{z}$  in the dataset  $D$ , as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

Regarding the approximation, the classical work by Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions  $v$  with integer values. More generally, their result yields an approximation factor of  $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$ . We showed in Proposition 2 that the value function  $v_{\text{con}}^g$  is monotone non-decreasing, and under feature independence, Proposition 4 establishes its submodularity. Combined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ .

However, we must also show that the expression  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$  is both finite and bounded by  $\mathcal{O}(\ln(|D|))$ , implying that it is effectively constant, due to the assumption of a fixed dataset  $|D|$ . To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator  $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$  to be zero. We begin by formally defining what we mean by redundancy:

**Definition 1.** *Let  $\mathcal{D}$  denote some empirical distribution over a dataset  $D$ . Then we say that some feature  $i \in [n]$  is redundant with respect to  $\mathcal{D}$  if for any pair  $\mathbf{x}, \mathbf{z} \in D$  it holds that  $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$ .*

Here, the notation  $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$  indicates that all features in  $[n] \setminus \{i\}$  are fixed to their values in  $\mathbf{x}$ , while feature  $i$  is set to its value in  $\mathbf{z}$ . Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation  $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$  indicates that the features in  $S$  are fixed to their values in  $\mathbf{x}$ , while the features in  $\bar{S}$  are fixed to their values in  $\mathbf{z}$ . Thus, in this sense, if we “remove” feature  $i$  from the input space and define a new function  $f'$  over the reduced space  $[n'] := [n] \setminus \{i\}$ , then for any input of size  $n'$ , the output of  $f'$  will exactly match the output of  $f$  when applied to the same features  $[n] \setminus \{i\}$ .

This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it involves detaching the corresponding input neuron from the network; for tree ensembles, it involves removing any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function  $v_{\text{con}}^g(S)$  — as previously discussed in this proof and in Proposition 2 — we can compute each  $v_{\text{con}}^g(\{i\})$  for all  $i \in [n]$  by iterating over all pairs  $\mathbf{x}, \mathbf{z} \in \mathcal{D}$  and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature  $i$  is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then  $i$  is deemed redundant. Once all redundant features have been identified with respect to the empirical distribution  $\mathcal{D}$ , we can remove them and construct an equivalent model  $f'$ . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that  $v_{\text{con}}^g(\{i\})$  equals 0 for some empirical distribution  $\mathcal{D}$  if and only if feature  $i$  is redundant with respect to  $\mathcal{D}$ . Therefore, once the preprocessing step removes all redundant features from  $f$  and we construct the resulting function  $f'$ , we ensure that  $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$ .

We now present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is  $\mathcal{O}(\ln(|\mathcal{D}|))$ . Given that the empirical distribution  $\mathcal{D}$  is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish that the bound is *constant* — a property that will later sharply contrast with the local explainability setting, where no bounded approximation exists.

More specifically, we know that the probability function  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$  is, by the definition of empirical distributions, at least  $\frac{1}{|\mathcal{D}|}$ . Therefore, the definition of  $\mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$  (i.e., the value function  $v_{\text{con}}^g$ ) is at least  $\frac{1}{|\mathcal{D}|^2}$ . In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|\mathcal{D}|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left( \frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|\mathcal{D}|^2) = \mathcal{O}(\ln(|\mathcal{D}|)) \quad (71)$$

Which concludes our proof.

□

**Lemma 11.** *Unless  $\text{PTIME} = \text{NP}$ , there is no bounded approximation for computing a cardinally minimal local  $\delta$ -contrastive reason for any  $\langle f, \mathbf{x} \rangle$  where  $f$  is either a neural network or a tree ensemble, even when  $|\mathcal{D}| = 1$ .*

*Proof.* Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local  $\delta$ -contrastive reason for some  $\langle f, \mathbf{x} \rangle$ , where  $f$  is a neural network or a tree ensemble—even when  $|\mathcal{D}| = 1$ . However, Lemma 9 establishes that even with a single baseline  $\mathbf{z} = 0_n$  (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless  $\text{PTIME} = \text{NP}$ . Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

## L PROOF OF THEOREM 4

**Theorem 4.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O} \left( \frac{1}{1-k_f} + \ln \left( \frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for*

computing a global cardinally minimal  $\delta$ -sufficient reason for  $f$ , assuming feature independence. In contrast, unless  $PTIME=NP$ , there is no bounded approximation for computing a local cardinally minimal  $\delta$ -sufficient reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .

*Proof.* We divide the proof into two lemmas: one establishing the approximation guarantee for the global case, and the other demonstrating the intractability of the local case.

**Lemma 12.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal  $\delta$ -sufficient reason for  $f$ , assuming feature independence.*

*Proof.* The proof will follow a similar approach to that of Lemma 10, where we showed that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of  $\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)$ . After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by  $\mathcal{O}(\ln(|D|))$ .

Similar to Lemma 10, we will again prove a stronger claim — namely, that the result holds for *any* model, assuming the trivial condition that its inference time is polynomially computable, and the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs  $\mathbf{x}, \mathbf{z}$  in the dataset  $D$ , as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, where the approximation relied on the result by Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the setting here requires a different condition due to the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) provided an approximation guarantee of  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ , where  $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$ .

We established in Proposition 2 that the value function  $v_{\text{suff}}^g$  is monotone non-decreasing, and under the assumption of feature independence, Proposition 3 further shows that it is supermodular. Given that Algorithm 2 runs in polynomial time, this leads directly to an approximation guarantee of  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ , where  $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$ .

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from  $f$  — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature  $i$  is strictly redundant if and only if  $v(\{i\}) - v(\emptyset) = 0$ . This preprocessing yields a new function  $f'$  that behaves identically to  $f$  over the remaining features and ensures that, for every  $i$ , both  $v_{\text{suff}}^g(\{i\}) > 0$  and  $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$  hold.

In order for us to show that this expression is both bounded and constant, similarly to Lemma 10 we will perform the exact same preprocessing phase (which, as we explained there, can be performed in polynomial time both for neural networks as well as tree ensembles) as before where we remove all redundant features from  $f$ . We note that here too, it holds that a feature  $i$  is strictly redundant if and only if  $v(\{i\}) - v(\emptyset) = 0$ . Hence, this preprocessing phase will give us a new function  $f'$  for which it both holds that it is equivalent to the behaviour of  $f$  for all remaining features and also satisfies that for any  $i$  it holds that both  $v_{\text{suff}}^g(\{i\}) > 0$  and  $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$  hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by  $\frac{1}{|D|}$  and its expected value by  $\frac{1}{|D|^2}$ , we obtain that  $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$  and  $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$ . This implies, as in Lemma 10, that the first term is lower bounded by  $\mathcal{O}(\ln(|D|))$ , and for the second term we derive the following:

Since  $v_{\text{suff}}^g$  is supermodular (Proposition 3), we have that  $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$ . It follows that  $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$ . This implies:  $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$ , and therefore,  $1 - |D|^2 \leq k^f \leq 0$ . This yields  $\frac{1}{1 - k^f} \leq |D|^2$ , demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly smaller in practice, our goal here is simply to show that it remains constant — unlike in the local setting, where no bounded approximation is achievable.

□

**Lemma 13.** *Unless  $\text{PTIME} = \text{NP}$ , there is no bounded approximation for computing a local cardinally minimal  $\delta$ -sufficient reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

*Proof.* The proof follows a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local  $\delta$ -sufficient reason for some  $\langle f, \mathbf{x} \rangle$ , where  $f$  is either a neural network or a tree ensemble—even when  $|D| = 1$ . Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline  $\mathbf{z} = 0_n$ , deciding whether a sufficient reason exists is NP-hard unless  $\text{PTIME} = \text{NP}$ . Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

## M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

**Proposition 1.** *Assume a neural network  $f : \mathbb{F} \rightarrow [c]$  with ReLU activations, some dataset  $\mathbf{D}$ , such that  $\mathcal{D}$  is the empirical distribution defined over  $\mathbf{D}$ , and some threshold  $\delta \in \mathbb{Q}$ . Then unless  $\text{NP} \subseteq \text{DTIME}(n^{\mathcal{O}(\log(\log(n)))})$ , there does not exist a polynomial-time  $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$  approximation algorithm for computing a cardinally minimal global contrastive reason for  $f$  concerning  $\mathcal{D}$ , for any fixed  $\epsilon > 0$ .*

*Proof.* The proof proceeds via a polynomial-time approximation-preserving reduction from a specific variant of Set Cover that is, similarly to regular Set Cover, provably hard to approximate within the factor given in the proposition (Vazirani, 2001).

### Partial Set Cover:

**Input:** A universe  $U := \{u_1, \dots, u_n\}$ , a collection of sets  $S := \{S_1, \dots, S_m\}$  such that for all  $j$   $S_j \subseteq U$ , and a coverage threshold  $\delta \in \{0, 1, \dots, n\}$ .

**Output:** A minimum-size subset  $C \subseteq S$  such that  $|\cup_{S_j \in C} S_j| \geq \delta$ .

We reduce this problem to that of computing a *global*, cardinally-minimal contrastive reason  $S$  for a neural network  $f$  with respect to a distribution  $\mathcal{D}$  defined over a dataset  $\mathbf{D}$ , and some threshold  $\delta'$ . We begin by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset  $\mathbf{D}$ . The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over  $\mathbf{D}$ , this implies that:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] = \\ & \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] = \\ & \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left( \frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) \end{aligned} \quad (74)$$

where we denote  $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$ . We now construct a ReLU neural network  $f$ . Before doing so, we specify the functional form that we intend the constructed neural network  $f$  to

implement. For each set  $S_j$ , we introduce two vectors  $\mathbf{x}^j$  and  $\mathbf{y}^j$ , both belonging to  $\mathbb{Z}^n$  and drawn from the dataset  $\mathbf{D}$ , such that they agree on all coordinates outside  $S_j$ ; that is, they share identical feature values on the complement  $\bar{S}_j$ . Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in  $S_j$  by selecting them from two distinct vectors  $\mathbf{z}$  and  $\mathbf{z}'$  such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^{\prime j}) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^{\prime j}) = \mathbf{y}^j \quad (76)$$

We aim to construct  $f$  such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^{\prime j}) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^{\prime j}) = f(\mathbf{y}^j) \quad (77)$$

Let us now assume that both  $\mathbf{x}^j$  and  $\mathbf{y}^j$  occur *uniquely* in the dataset  $\mathbf{D}$  across all  $[n]$  coordinates — that is, no other vector in  $\mathbf{D}$  matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{S_j} = \mathbf{x}_{S_j})] = \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left( \frac{\sum_{\mathbf{z} \in \mathbf{D}_{S_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{S_j}(\mathbf{z}, \mathbf{x})} \right) = \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs —  $(\mathbf{x}^j, \mathbf{z}^j)$  and  $(\mathbf{z}^j, \mathbf{x}^j)$  — contribute to the respective sums.

We repeat this construction for each  $S_i$  in the set-cover instance, each time choosing two *distinct* vectors  $\mathbf{x}^i$  and  $\mathbf{y}^i$  that differ on *every* coordinate in  $[n]$ . In total, this yields  $2m$  vectors that form the dataset  $\mathbf{D}$  from which we construct  $f$ .

We then construct a neural network  $f$  that simulates this function. This is achieved using a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network capable of representing any finite set of input–output pairs can be built in polynomial time. Using this result, we construct  $f$  accordingly, and set  $\delta' := \frac{\delta}{|\mathbf{D}|}$ .

We now argue that a subset  $S_j$  is a global contrastive reason for  $f$  under tolerance  $\delta'$  if and only if  $C$  is a partial set cover for  $U$  under tolerance  $\delta$ . By Equation 78, each constructed pair of vectors  $(\mathbf{x}^j, \mathbf{y}^j)$  contributes exactly  $\frac{1}{|\mathbf{D}|}$  to the value term. Moreover, because each  $S_j$  uses its own distinct pair of vectors in  $\mathbb{Z}^n$ , there is no overlap between these pairs. Consequently, the coverage achieved by any set cover  $C$  corresponds exactly to the accumulated contribution of the selected subsets in the input domain of  $f$ . This completes the reduction.

□

## N TRACTABILITY RESULTS FOR ORTHOGONAL DNFS

**Lemma 14.** *If  $f$  is an orthogonal DNFs and the probability term  $v_{\text{suff}}^g$  can be computed in polynomial time given the distribution  $\mathcal{D}$  (which holds for independent distributions, among others), then obtaining a subset-minimal global  $\delta$ -sufficient reason can be obtained in polynomial time.*

*Proof.* We begin by defining orthogonal DNFs (Crama & Hammer, 2011) before presenting the full proof.

**Definition 2** (Orthogonal DNFs). *A Boolean function  $\varphi$  is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms  $T_1 \vee T_2 \vee \dots \vee T_m$ , such that for every pair of distinct terms  $T_i$  and  $T_j$  ( $i \neq j$ ):*

$$T_i \wedge T_j \models \perp$$

*This means that no single variable assignment can satisfy more than one term simultaneously.*

2214 The proof mirrors the argument used in Theorem 1, relying on the fact that  $v_{\text{suff}}^g$  can be computed in  
 2215 polynomial time for any set  $S$ .

2216 For a given set  $S$ , let  $\mathcal{C}^S T_i$  be the constraints that the term  $T_i$  imposes on the variables in  $S$ .  
 2217 We enumerate all pairs of terms  $(T_i, T_j)$ . Any pair whose constraints on  $S$  are *inconsistent* (i.e.,  
 2218  $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$ ) is ignored, since it contributes zero to  $v_{\text{suff}}^g$ .  
 2219

2220 For each *consistent* pair of terms  $(T_i, T_j)$ , we evaluate  $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$ , where  $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$ .  
 2221 The contribution of this pair to  $v_{\text{suff}}^g$  is then  
 2222

$$2223 \Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

2224  
 2225 Under feature independence, these probabilities factor as  $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$ . Since there are at  
 2226 most  $m^2$  such term pairs, the value of  $v_{\text{suff}}^g$  can be computed in polynomial time under an independent  
 2227 distribution  $\mathcal{D}$ .  $\square$   
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## 2229 O DISCLOSURE: USAGE OF LLMS

2230  
 2231 An LLM was used exclusively as a writing assistant to refine grammar and typos and improve clarity.  
 2232 It did not contribute to the generation of research ideas, study design, data analysis, or interpretation  
 2233 of results, all of which were carried out solely by the authors.  
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